Domain Specific Language Design (2IMP20): Syntax

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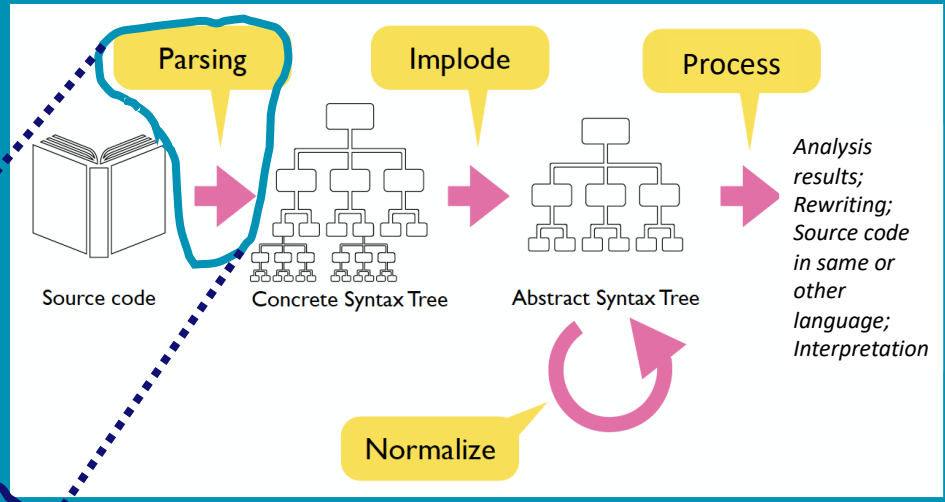
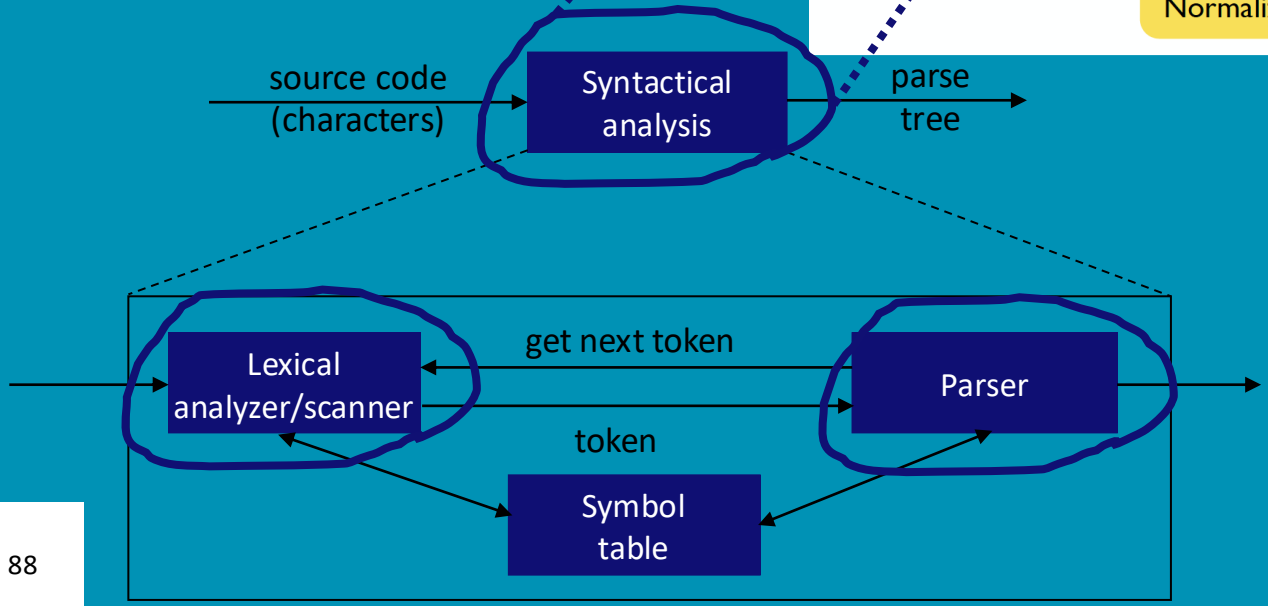
Lexical and syntactical analysis using LEX and YACC

Recall: Scanning/lexical analysis + parsing/syntactical analysis





… but why this old stuff??

- FORTRAN,
- COBOL,
- LEX and YACC

instead of DSLs and Rascal/EMF/Xtext/ ANTLR/Langium…

(souce: https://www.nrc.nl/nieuws/2024/05/01/gezocht…)

Lexical and syntactical analysis using LEX and YACC

- YACC: Yet Another Compiler Compiler
- LEX and YACC work as a team

- Old: 1960s/1970s…
- *originally…*
- *extensively used*
- *highly available*
  - **lex**, **yacc** on most UNIX systems
  - **bison**: a yacc replacement from GNU
  - **flex**: fast lexical analyzer
  - BSD yacc
  - Versions for other OS'es (MS Windows, Linux, …)

Lexical analysis using (F)LEX

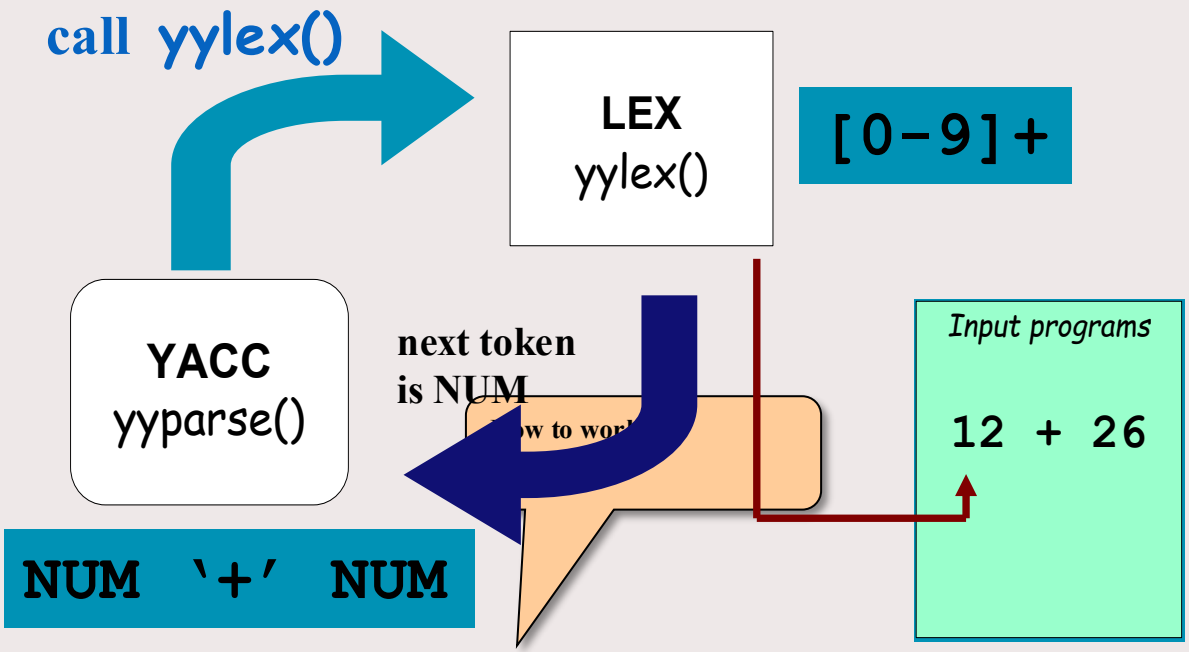
LEX allows:

- character classes, e.g. [0-9]+ in regular expressions
- "semantic" actions with regular expression
  - `[0-9]+ { value = atoi(yytext); sum += value;}`
  - `yytext` contains the recognized *string (token)*
  - in accepting state (i.e. when regex is matched) the semantic action is executed

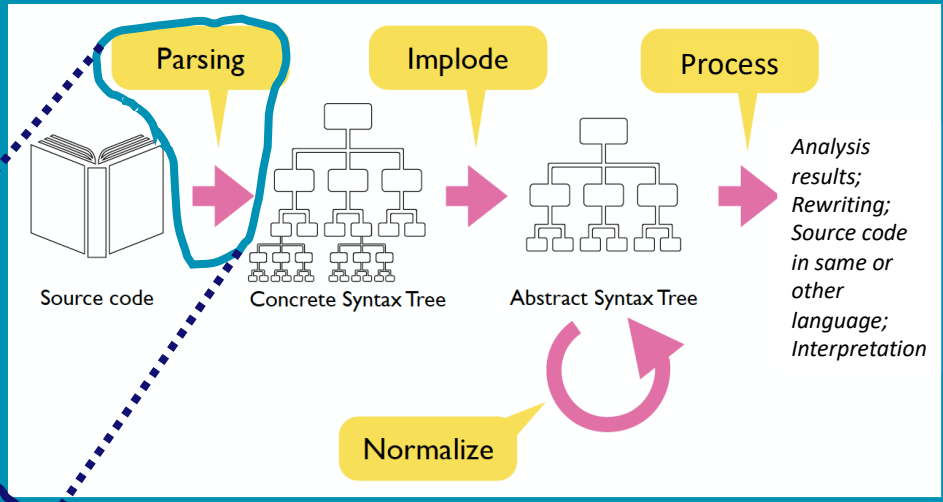
Example of a (F)LEX specification

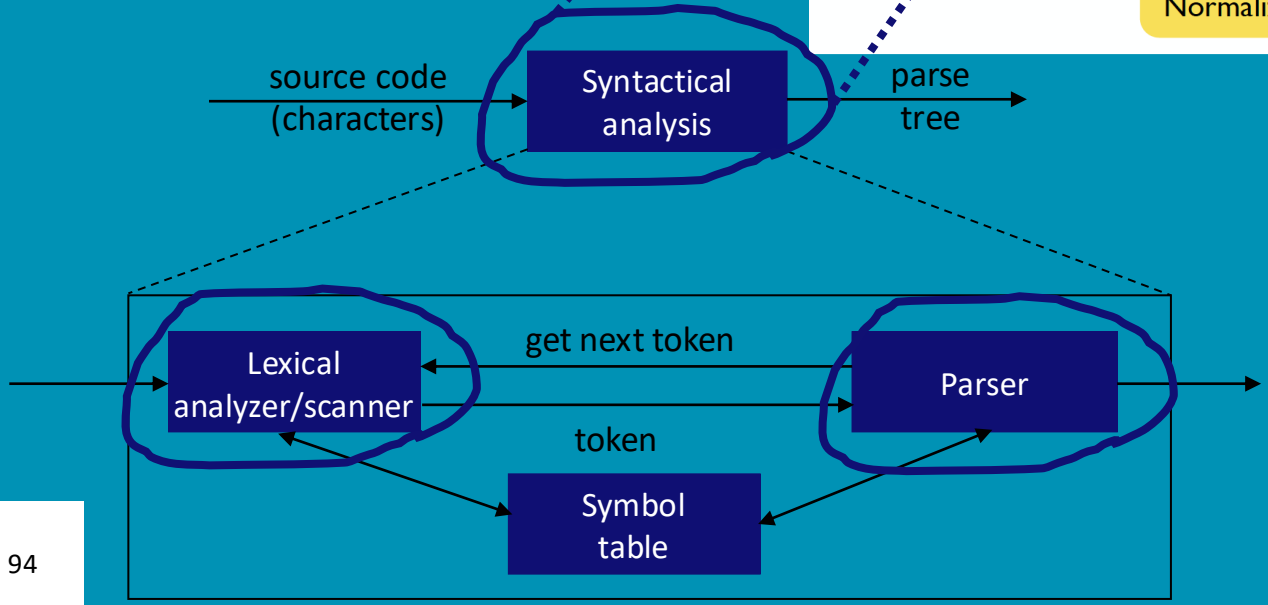
```
/*** Definition section ***/
%{ /* C code to be copied verbatim */
   #include <stdio.h>
%}
/* This tells flex to read only one input file */
%option noyywrap
%%
/*** Rules section ***/
/* [0-9]+ matches a string of one or more digits */
[0-9]+ { /* yytext is a string containing the matched text. */
  printf("Saw an integer: %s\n", yytext);
}
.|\n { /* Ignore all other characters. */ }
%%
/*** C Code section ***/
int main(void) {
/* Call the lexer, then quit. */
  yylex();  return 0; }
```

LEX and YACC

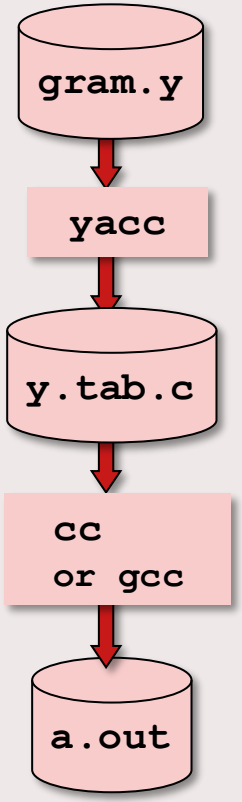

call yylex()
LEX
yylex()
[0-9]+
YACC
yyparse()
next token
is NUM
Input programs
12 + 26
NUM '+' NUM

Recall: Scanning/lexical analysis + parsing/syntactical analysis





Syntactical analysis using YACC



File containing desired grammar in YACC format

YACC program

C source program created by YACC

C compiler

Executable program that will parse programs conforming to grammar given in `gram.y`

Syntactical analysis using YACC – .y File Structure

```
Definitions

%%

Rules

%%

Supplementary Code
```

The identical LEX format was taken from this…

Syntactical analysis using YACC

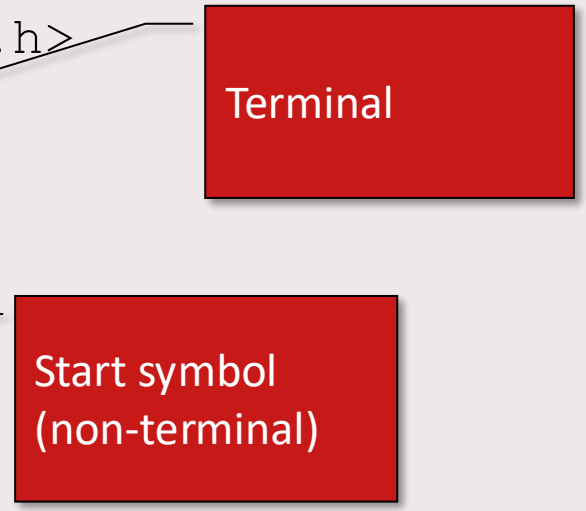
Rules section: a context free grammar, e.g.:

```
expr   : expr '+' term
       | term
       ;
term   : term '*' factor
       | factor
       ;
factor : '(' expr ')'
       | ID
       | NUM
       ;
```

Syntactical analysis using YACC

Definition section

```
%{
#include <stdio.h>
#include <stdlib.h>
%}
%token ID NUM
%start expr
```



Syntactical analysis using YACC

Some details

- LEX produces a function called *yylex*() — *scanner/tokenizer*
- YACC produces a function called *yyparse*() — *parser*

- *yyparse*() expects to be able to call *yylex*()
  How to get *yylex*()?
  1. If you don't want to write your own: use lex!
  2. Write your own!

Syntactical analysis using YACC

Semantic actions

```
expr : expr '+' term     { $$ = $1 + $3; }
     | term              { $$ = $1; }
     ;
term : term '*' factor   { $$ = $1 * $3; }
     | factor            { $$ = $1; }
     ;
factor : '(' expr ')'    { $$ = $2; }
       | ID
       | NUM
       ;
```

Syntactical analysis using YACC

Semantic actions

```
expr : expr '+' term     { $$ = $1 + $3; }
     | term              { $$ = $1; }
     ;
term : term '*' factor   { $$ = $1 * $3; }
     | factor            { $$ = $1; }
     ;
factor : '(' expr ')'    { $$ = $2; }
       | ID
       | NUM
       ;
```

Syntactical analysis using YACC

Semantic actions

$1

```
expr : expr '+' term     { $$ = $1 + $3; }
     | term              { $$ = $1; }
     ;
term : term '*' factor   { $$ = $1 * $3; }
     | factor            { $$ = $1; }
     ;
factor : '(' expr ')'    { $$ = $2; }
       | ID
       | NUM
       ;
```

Syntactical analysis using YACC

Semantic actions

```
expr : expr '+' term     { $$ = $1 + $3; }
     | term              { $$ = $1; }
     ;
term : term '*' factor   { $$ = $1 * $3; }
     | factor            { $$ = $1; }
     ;
factor : '(' expr ')'    { $$ = $2; }
       | ID
       | NUM

       ;
```

$2

Syntactical analysis using YACC

Semantic actions

```
expr : expr '+' term     { $$ = $1 + $3; }
     | term              { $$ = $1; }
     ;
term : term '*' factor   { $$ = $1 * $3; }
     | factor            { $$ = $1; }
     ;
factor : '(' expr ')'    { $$ = $2; }
       | ID
       | NUM
       ;
```

$3

Syntactical analysis using YACC – Precedence and association

Precedence/association

```
expr: expr '-' expr
    | expr '*' expr
    | expr '<' expr
    | '(' expr ')'
     ...
    ;
```

(1) 1 – 2 – 3

(2) 1 – 2 * 3

1. 1-2-3 = (1-2)-3? or 1-(2-3)?
   Define '-' operator with left-association.
2. 1-2*3 = 1-(2*3)
   Define "*" operator as precedent to "-" operator

Syntactical analysis using YACC – Precedence and association

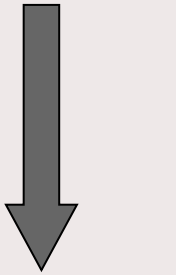
Precedence/association

```
%left  '+' '-'
%left  '*' '/'
%noassoc UMINUS
expr   : expr '+' expr      { $$ = $1 + $3; }
       | expr '-' expr      { $$ = $1 - $3; }
       | expr '*' expr      { $$ = $1 * $3; }
       | expr '/' expr      { if($3==0)
                                 yyerror("divide 0");
                              else
                                 $$ = $1 / $3;
                            }
       | '-' expr %prec UMINUS {$$ = -$2; }
```

Syntactical analysis using YACC – Precedence and association

Precedence/association

```
%right '='
%left  '<' '>' NE LE GE
%left  '+' '-'
%left  '*' '/'
```



highest precedence

Syntactical analysis using YACC

YACC

- Rules may be recursive
- Rules may be ambiguous
- Uses *bottom-up Shift/Reduce parsing*
  - Get a token
  - Push onto stack
  - Can it be reduced (How do we know?)
    - If yes: Reduce using a rule
    - If no: Get another token
- YACC cannot look ahead > 1 token

Syntactical analysis using YACC – Conflicts

Shift/reduce conflict

occurs when a grammar is written in such a way that a decision between shifting and reducing can not be made.

e.g.: IF-ELSE ambiguity

- To resolve conflict, *YACC will choose to shift*

Reduce/reduce conflict

```
start : expr | stmt
              ;
expr : CONSTANT;
stmt : CONSTANT;
```

- To resolve conflict, *YACC will reduce using earlier grammar rule*

Not good practice to rely on this → modify grammar to eliminate them

Syntactical analysis using YACC – Use left recursion

Left recursion

```
list:
      item
    | list ',' item
    ;
```

Right recursion

```
list:
      item
    | item ',' list
    ;
```

LR parsers prefer left recursion
LL parsers prefer right recursion
YACC is an LR parser

Lexical and syntactical analysis using LEX and YACC

Further reading on LEX+YACC:
http://dinosaur.compilertools.net
http://flex.sourceforge.net/manual/
http://www.delorie.com/gnu/docs/bison/bison_toc.html

Questions?